Computer Network report
Semester: VI

SECURE NETWORK IMPLEMENTATION USING VPN, FIREWALL AND IPS

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1. Abstract

This project focuses on designing and implementing a secure network architecture using key security technologies such as Virtual Private Network (VPN), Firewall, and Intrusion Prevention System (IPS). The main objective is to protect an organization's network from unauthorized access and cyber threats while ensuring secure communication.

A network topology was created using Cisco Packet Tracer, consisting of internal devices, a router, firewall, server, and an external attacker system. The router was configured to act as a VPN gateway, while firewall rules and Access Control Lists (ACLs) were applied to control traffic and simulate IPS functionality.

The results demonstrate that the configured system successfully blocks unauthorized access while allowing legitimate communication, ensuring a secure and efficient network environment.

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2. Introduction

Objective

To design a secure network system that protects internal resources using VPN, firewall policies, and intrusion prevention techniques.

Problem Statement

In today's digital environment, networks are constantly exposed to threats such as hacking, malware, and unauthorized access. Organizations require a robust security mechanism to ensure safe communication and data protection. This project addresses the need for implementing layered security using VPN, firewall, and IPS techniques.

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3. System Analysis & Design

Requirements

Hardware:

- Computer/Laptop

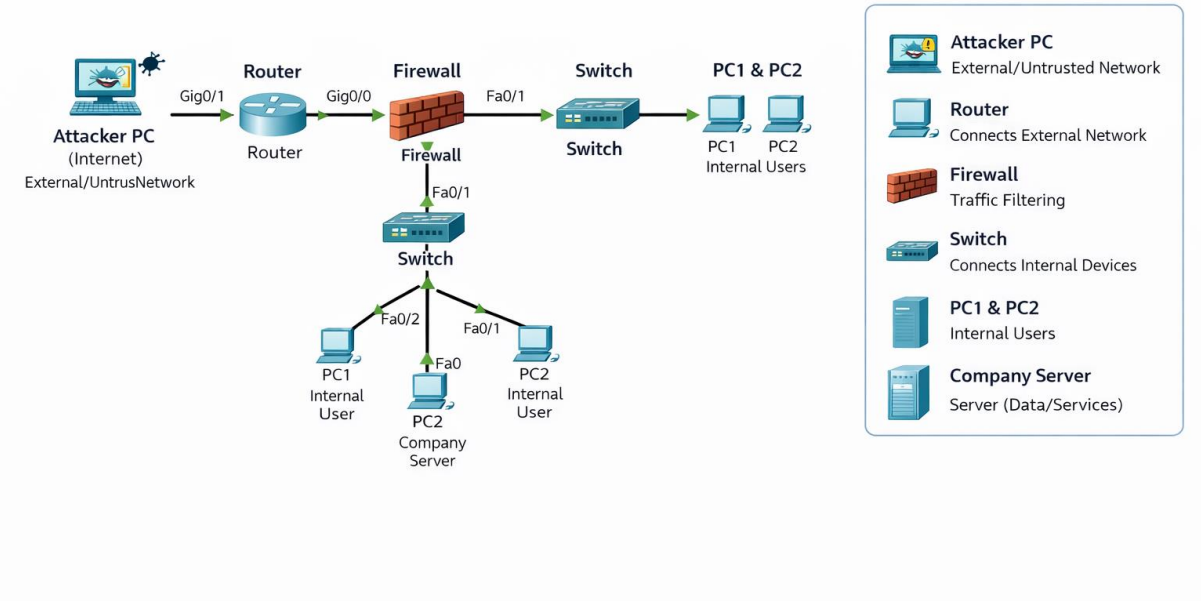
Software:

- Cisco Packet Tracer

Knowledge Required:

- Basic networking concepts
- IP addressing
- Routing and security fundamentals

Block Diagram (Concept)



- Router acts as VPN Gateway
- Firewall controls incoming and outgoing traffic
- ACLs simulate IPS functionality

- Internal network connected to server and PCs
- External attacker connected via internet side

Methodology

- Designed network topology using Cisco Packet Tracer
- Assigned IP addresses to all devices
- Connected internal and external networks
- Configured firewall for traffic filtering
- Applied ACL rules on router
- Tested connectivity and security using ping commands

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4. Implementation

4.1 IP Configuration Module

- Configured router interfaces:
  - Outside Interface: `200.1.1.1`
  - Inside Interface: `192.168.1.1`
- Assigned IPs to PCs and server:
  - PC1: `192.168.1.10`
  - PC2: `192.168.1.11`
  - Server: `192.168.1.100`
- Configured attacker PC:
  - `200.1.1.2`

4.2 Router Configuration

```
enable
configure terminal

interface g0/1
ip address 200.1.1.1 255.255.255.0
no shutdown

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

4.3 Firewall Configuration

```
enable
configure terminal
```

```
interface vlan 1
nameif inside
security-level 100
ip address 192.168.1.254 255.255.255.0

interface vlan 2
nameif outside
security-level 0
ip address 200.1.1.3 255.255.255.0
```

4.4 Security Configuration (ACL)

- Created ACL to block attacker IP
- Applied ACL on router interface
- Ensured restricted access to internal network

4.5 VPN Concept

- Router configured as VPN Gateway
- Enables secure remote access (conceptual implementation)

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5. Testing and Results

Testing Performed

- PC1 to Server communication → Successful
- PC to Router connectivity → Verified
- Attacker to Router → Reachable
- Attacker to Internal Network → Blocked

Observations

- Internal network communication works properly
- External attacker is unable to access internal devices
- Firewall and ACL effectively protect network

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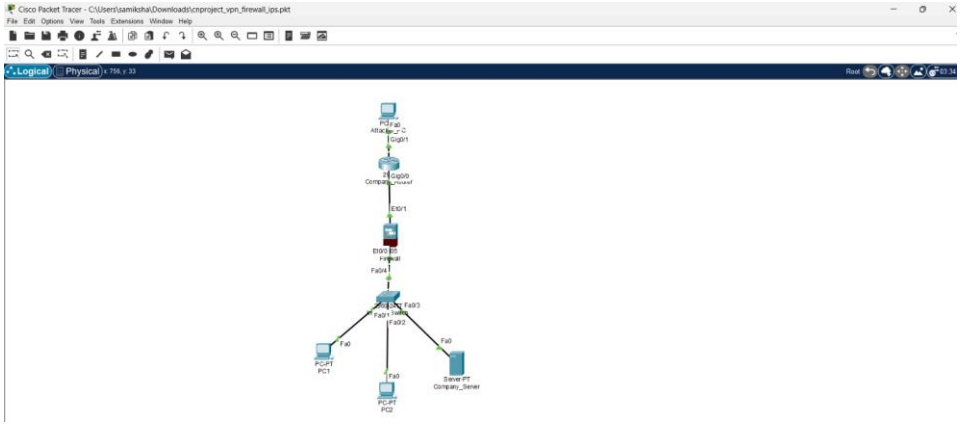
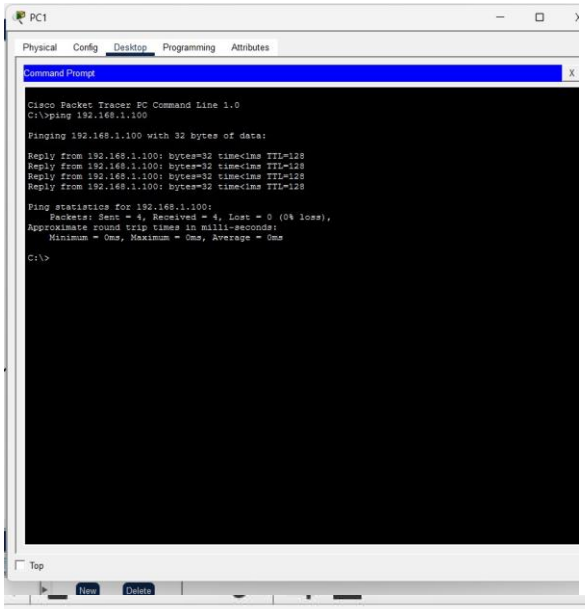
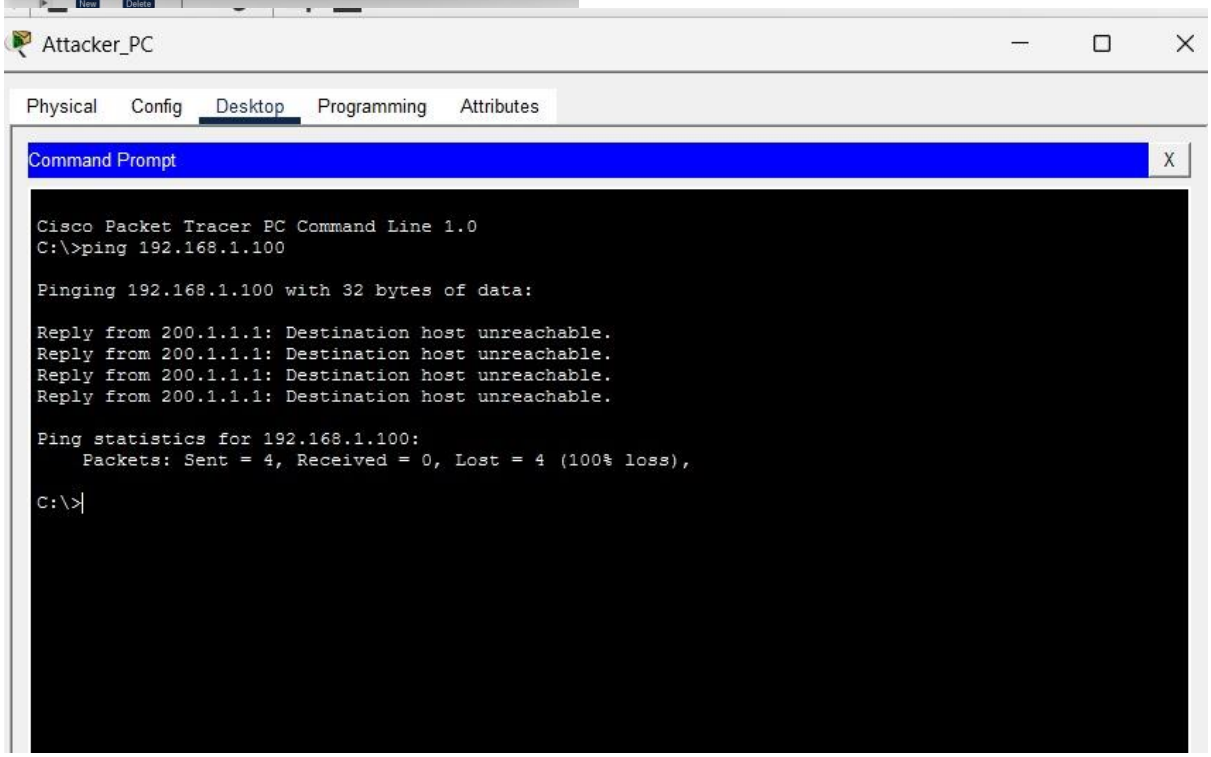
6. Results and Discussion




PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.100
Pinging 192.168.1.100 with 32 bytes of data:
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.100:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top


Attacker_PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.100
Pinging 192.168.1.100 with 32 bytes of data:
Reply from 200.1.1.1: Destination host unreachable.
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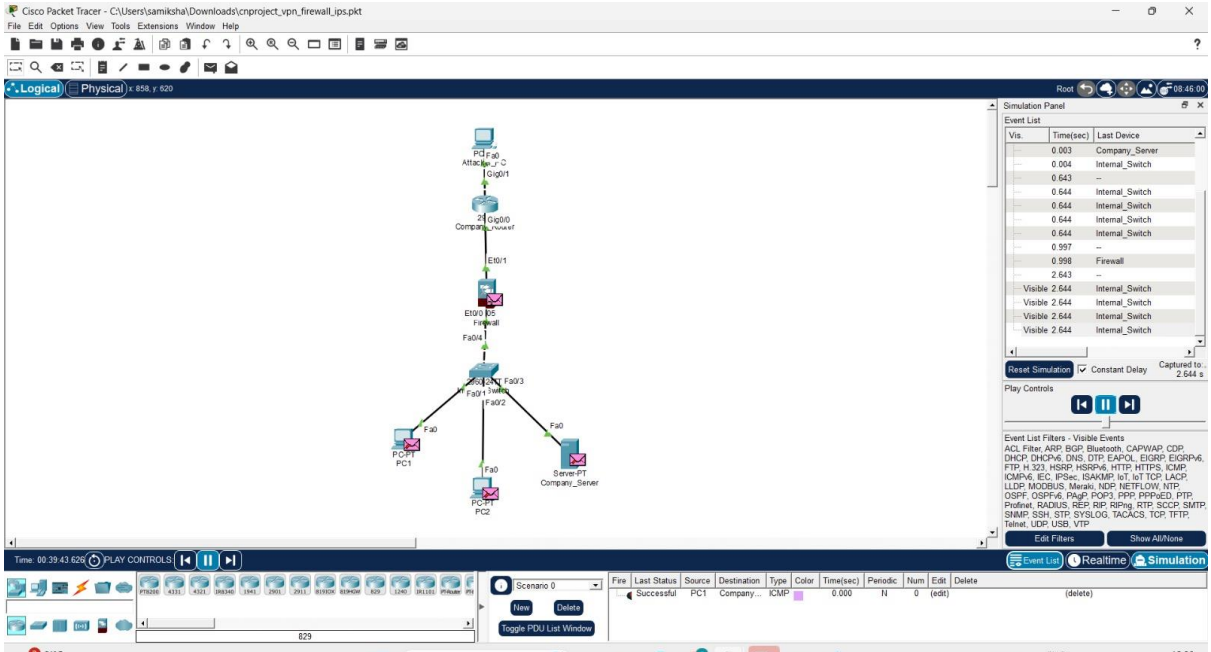
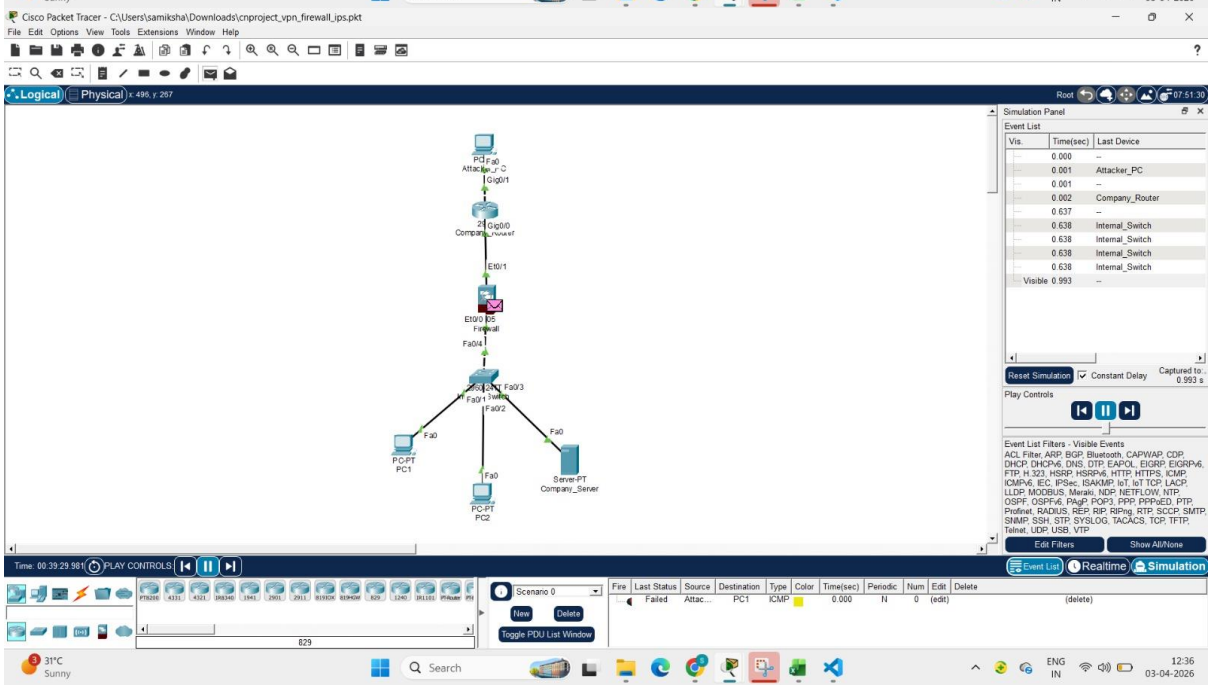
Ping statistics for 192.168.1.100:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>





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7. Conclusion & Future Scope

Conclusion

The project successfully implements a secure network using VPN, firewall, and IPS concepts. The firewall and ACL mechanisms effectively block unauthorized access while maintaining smooth internal communication. The use of layered security ensures confidentiality, integrity, and availability of network resources.

Future Scope

- Implement real VPN configuration (IPSec)
- Integrate advanced IDS/IPS systems
- Add network monitoring tools
- Deploy cloud-based security solutions

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8. References

- Cisco Packet Tracer Documentation
- Computer Networks – Andrew S. Tanenbaum
- Cisco Official Website
- Networking Tutorials (GeeksforGeeks)

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